Fall 2025.

Physics 501. Methods of Theoretical Physics I.

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Homework # 3.

Due to October 7^{th}, 2025.

Read the corresponding chapters. If you forgot anything necessary from your core math courses, find the information in the textbook, online or whatever works for you. Solve the problems. It should be done in a very detailed and logical way. You need to understand every step of your solution.

Vector Analysis.

1. **Arfken&Weber, 7th Edition. Chapter 3. Read it.**

2. **Prove the identities:**

a) $curl(\varphi\vec{A}) = \varphi curl\vec{A} - \vec{A} \times grad\varphi;$

b) $curl(\vec{A} \times \vec{B}) = \vec{A}\, div\vec{B} - \vec{B}\, div\vec{A} + (\vec{B} \cdot \nabla)\vec{A} - (\vec{A} \cdot \nabla)\vec{B};$

c) $(\nabla \times \vec{A}) \times \vec{B} = \vec{A}\, div\vec{B} - (\vec{A} \cdot \nabla)\vec{B} - \vec{A} \times curl\, \vec{B} - \vec{B} \times curl\, \vec{A};$

d) $(\vec{A} \times \vec{B}) \cdot (\vec{C} \times \vec{D}) + (\vec{B} \times \vec{C}) \cdot (\vec{A} \times \vec{D}) + (\vec{C} \times \vec{A}) \cdot (\vec{B} \times \vec{D}) = 0.$

3. Calculate $curl(\vec{\omega} \times \vec{r})$ and $div(\vec{\omega} \times \vec{r})$ where $\vec{\omega}$ is a constant vector.

4. The two-dimensional parabolic system of coordinates is defined as following:

$$x = \xi\eta,\ y = \frac{1}{2}(\xi^2 - \eta^2)$$

Find Lame coefficients and the Laplacian $\nabla^2 f$ in this system of coordinates. Do not forget to double-check if the system is orthogonal.

5. The bispherical system of coordinates is defined as following:

$$x = \frac{a\ \sin\xi\ \cos\phi}{\cosh\eta - \cos\xi};\ y = \frac{a\ \sin\xi\ \sin\phi}{\cosh\eta - \cos\xi};\ z = \frac{a\ \sinh\eta}{\cosh\eta - \cos\xi}$$

Find Lame coefficients and the Laplacian $\nabla^2 f$ in this system of coordinates. Do not forget to double-check if the system is orthogonal.

6. Calculate components of Laplacian of a vector function, $\nabla^2 \vec{A}$, in cylindrical system of coordinates. Use the formula I proved in the classroom: $curl\ curl\ \vec{A} = grad\ div\ \vec{A} - \nabla^2 \vec{A}$ and expressions for vector differential operations in cylindrical system of coordinates from the Arfken&Weber textbook.

The answer:

$$(\nabla^2 \vec{A})_r = \frac{\partial^2 A_r}{\partial r^2} + \frac{1}{r}\frac{\partial A_r}{\partial r} + \frac{1}{r^2}\frac{\partial^2 A_r}{\partial \varphi^2} + \frac{\partial^2 A_r}{\partial z^2} - \frac{2}{r^2}\frac{\partial A_\varphi}{\partial \varphi} - \frac{A_r}{r^2}$$

$$(\nabla^2 \vec{A})_\varphi = \frac{\partial^2 A_\varphi}{\partial r^2} + \frac{1}{r}\frac{\partial A_\varphi}{\partial r} + \frac{1}{r^2}\frac{\partial^2 A_\varphi}{\partial \varphi^2} + \frac{\partial^2 A_\varphi}{\partial z^2} + \frac{2}{r^2}\frac{\partial A_r}{\partial \varphi} - \frac{A_\varphi}{r^2}$$

$$(\nabla^2 \vec{A})_z = \frac{\partial^2 A_z}{\partial r^2} + \frac{1}{r}\frac{\partial A_z}{\partial r} + \frac{1}{r^2}\frac{\partial^2 A_z}{\partial \varphi^2} + \frac{\partial^2 A_z}{\partial z^2}$$

Tensor Algebra.

1. Represent the product $\left[\vec{a}\cdot\left(\vec{b}\times\vec{c}\right)\right][\vec{m}\cdot(\vec{n}\times\vec{p})]$ as a sum of such terms, which contain scalar (dot) products of vectors only.

2. Assume that $\vec{n}$ is a unit vector and all its directions are equiprobable, i.e., the system is isotropic. Calculate average values over all possible directions, i.e., over the solid angle of the following quantities, where $\vec{a}$, and $\vec{b}$, are constant vectors: $\overline{(\vec{a}\cdot\vec{n})^2}$; $\overline{(\vec{a}\times\vec{n})^2}$; $\overline{(\vec{a}\times\vec{n})\cdot\left(\vec{b}\times\vec{n}\right)}$.

3. Use the formulae (3.35) from the Arfken&Weber textbook to derive the formula (3.37).

4. Prove that a partial derivative $\frac{\partial A_i}{\partial x_k}$ of a vector A_i is a second rank tensor.

5. Show that a symmetry or asymmetry of a rotational tensor is invariant under rotation, i.e., if a tensor is (a)symmetric in one reference frame, it is still (a)symmetric for any new reference frame that is the result of rotation of the original one.